



**HKU
Med**

School of Clinical Medicine
Department of Medicine
香港大學內科學系



**HKU
Immunology &
Allergy**

Why do I always get sick?

Introduction to immunodeficiencies

Dr Philip H Li

**Division of Rheumatology & Clinical Immunology,
Department of Medicine, The University of Hong Kong**

Case Presentation

- M/22, salesperson
- Referred from a regional hospital for *?possible immunodeficiency*

HPI:

- Previous good past health until 2 years ago
- First admitted 2 years ago for fever, neck stiffness and headache
- **Not responsive to an empirical course of oral antibiotics**
- CT brain unremarkable → lumbar puncture performed

First episode – Sept 2017

Collect Date	27/09/17 16:46	Collect Date	27/09/17 21:21	27/09/17 21:21	27/09/17 21:21	Collect Date	27/09/17 21:21
Haemoglobin, Blood	17.1	Type, Specimen	CSF	CSF	CSF	Protein, CSF	0.76 ↑
WBC	19.9 ↑	Smear, Fungus	---	Smear negative	---	Glucose, CSF	3.3
Platelet	218	Gram Stain, CSF	---	---	No organism seen		
MCV	88.7	WBC, CSF	715	---	---		
MCH	31.0	RBC, CSF	36	---	---		
MCHC	35.0	Polymorph, CSF	77	---	---		
RBC	5.5	Lymphocyte, CSF	23	---	---		
HCT	0.49	Culture, Blood	---	---	---		
RDW	12.0	Culture, CSF	---	---	No growth		
MPV	11.0	Appearance, CSF	Clear	---	Colour:Yellowi sh Clear		
Neutrophil, absolute	16.6 ↑						
Neutrophil, %	83.8						

- Diagnosed with **bacterial meningitis** – responded to course of IV ceftriaxone
- *?False negative cultures due to empirical course of ceftriaxone*

Second episode – Jan 2018

- One year later, presented again with similar symptoms (fever + meningism)
- Complicated by septic shock

Collect Date	25/01/18 11:37
Haemoglobin, Blood	18.1 ↑
WBC	21.0 ↑
Platelet	256
MCV	92.2
MCH	31.3
MCHC	33.9
RBC	5.79
HCT	0.53 ↑
RDW	12.8
MPV	8.5

Collect Date	26/01/18 19:01	26/01/18 19:01	25/01/18 11:37
Type, Specimen	CSF	CSF	Blood
Smear, AFB	---	---	---
Gram Stain, Blood	---	---	Gram negative diplococci present
Gram Stain, CSF	---	---	---
WBC, CSF	5	---	---
WBC, Sputum	---	---	---
RBC, CSF	36	---	---
Polymorphs %, Fluid	58	---	---
Lymphocytes %, Fluid	42	---	---

Specimen Type:	Blood
Nature of Specimen:	Culture isolate
Culture:	
Organism 1:	Neisseria meningitidis

Both blood and CSF cultures yielded Neisseria meningitidis serotype Y

- Responded to ceftriaxone but ***complicated by left-sided hearing deficit***

Third episode – Jun 2018

- Six months later, admitted again for a third episode of meningitis

Collect Date	13/06/18 02:19
Haemoglobin, Blood	17.6 ↑
WBC	16.8 ↑
Platelet	291
MCV	88.8
MCH	30.8
MCHC	34.7
RBC	5.7
HCT	0.51 ↑
RDW	11.7
MPV	9.8
Neutrophil, absolute	15.9 ↑

Gram stain :-

Gram negative cocci seen

Culture :-

Organism 1 : *Neisseria meningitidis*

***Both blood and CSF cultures yielded
Neisseria meningitidis serotype Y***

- CT & MRI brain, sinuses and base of skull unremarkable.
- **ENT & Neurology: No structural abnormalities or cause found**
- **Other non-immunological secondary causes excluded**

Waiting for the fourth episode?

 **Department of Health**
The Government of the Hong Kong Special Administrative Region

Notifications and Deaths of Notifiable Infectious Diseases in 2018@

Disease	Notification		Death	
	Number	Rate [#]	Number	Rate [#]
Meningococcal infection (invasive)	10	0.13	1	0.01

Contributed to **3/10** cases of HK in 2018 ...



➔ *Referred to QMH Immunology Clinic*

Referred to QMH Immunology Clinic



Past Medical History: Unremarkable

- No history of frequent infections
- Received all vaccinations according to the immunisation programme
- No history of STIs, including *Neisseria gonorrhoeae*

Social History:

- Smoker and social drinker
- Working as a salesperson
- Born and raised in Hong Kong



Family History

- Youngest of six siblings (3M, 3F)
- No consanguinity
- Mother no history of miscarriages
- Father died in road traffic accident, unremarkable past health
- Both parents and six siblings were asymptomatic
- No family history of recurrent infections or meningococcal disease

First-line screening tests

- Complete blood count, differential counts and peripheral smear
- Liver and renal function tests
- Immunoglobulin G/A/M/E + electrophoresis
- C3, C4, ANA, anti-ENA ± other autoimmune serology (if indicated)
- Erythrocyte sedimentation rate, C-reactive protein
- HIV status
- CXR, CT thorax, pulmonary function tests

Second-line immunological tests

- Lymphocyte subsets: CD3, CD4, CD8, CD16/56, CD19
- Cell mediated immunity assessment:
 - Lymphocyte proliferation responses to mitogen / antigen
- Humoral immune response assessment:
 - Vaccine titres, e.g. post-pneumococcal vaccine serology
 - Coomb's test and isohemagglutinins (anti-A, anti-B)
 - IgG subclass, IgD and specific immunoglobulin levels
 - B cell immunophenotyping
- Complement function tests: CH50 / AH50
- C1-esterase inhibitor level and function
- Neutrophil function tests (e.g. NBT, DHR)
- Specific molecular/genetics tests & next-generation sequencing

HANDBOOK of INTERNAL MEDICINE

**COC(Medicine)
Hospital Authority**

**8th Edition
2019**

Immunology and Allergy

Anaphylaxis	IA 1-2
Drug Allergy	IA 3-4
Penicillin Allergy	IA 5
Angioedema	IA 6-7
Immunodeficiency	IA 8-11

Initial investigations

- Complete blood count: Unremarkable
- Absolute lymphocyte count and other differentials: Normal
- Liver and renal function tests: Normal
- HIV antibody/antigen negative
- Anti-HBs and anti-A/-B isohemagglutinins present
- CXR: Unremarkable

Hospital Code	QMH
Collect Date	23/08/18
Complement C3	81
Complement C4	13
C-Reactive Protein	<0.35
IgA	206
IgG	1230
IgM	207
Protein Electrophoresis	No monoclonal Ig detected.

Hospital Code	QMH
Collect Date	10/10/18 12:04
Type, Specimen	EDTA blood
NK Cells, CD16/56, %	5.5 ↓
CD3	2590 ↑
CD8, %	24.1
CD19, %	17.1
CD19	581 ↑
CD3, %	77.5
CD4, %	47.1
CD4	1546 ↑
CD8	792
CD4/CD8, Ratio	1.95
NK Cells, CD16/56	187

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← Order only based on
clinical suspicion!

Complement function

Division of Clinical Immunology

Rm 004 14/F Block T, 122 Pokfulam Road, Hong Kong
Tel. 22552327 Fax. 28181104

Sex/Age: M/22Y

Req. Loc.: QMH/MED/%RH

Doctor: DR. LI PHILIP HEI

Reference: IMMUNOLOGY CLINIC S6

Date Collected: 23/08/2018 --:--

Reference Interval

Date Arrived: 24/08/2018 --:--

Clinical Details: RECURRENT NEISURA(REFER TO REQUEST FORM).

CH50 Classical Pathway :

Control	22.0	CH50 units/ml	20 - 40
---------	------	---------------	---------

Patient	See Below	CH50 units/ml	20 - 40
---------	-----------	---------------	---------

Undetectable CH50 & AH50 activity observed. Please consider to repeat the assay to confirm the findings. If the abnormality persists upon repeated testing, possibility of deficiency in C3 or terminal pathway components (C5, C6, C7, C8 or C9) needs to be considered.

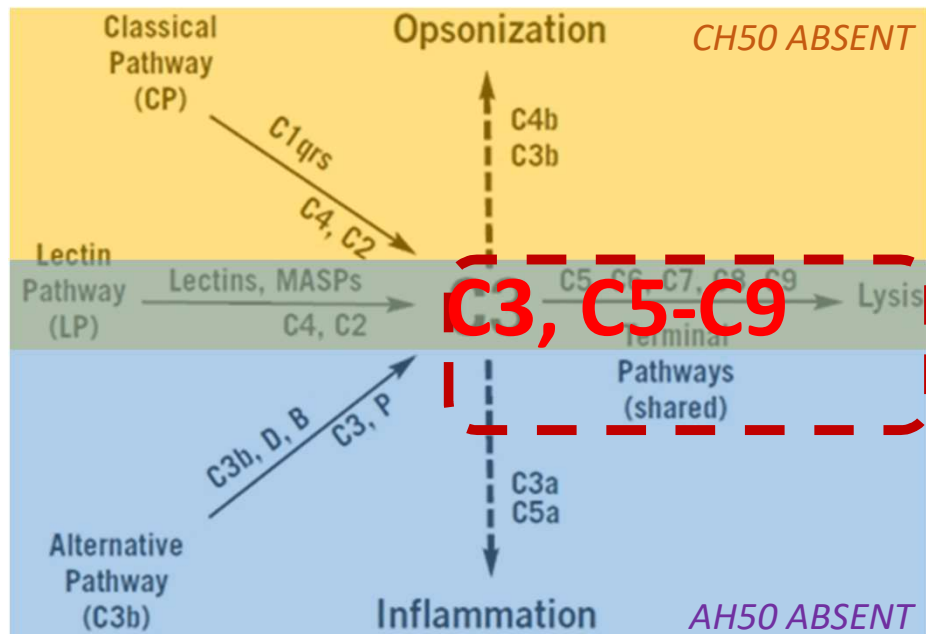
CH50 Alternative Pathway :

Control	28.0	CH50 units/ml	25 - 45
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Patient	See Below	CH50 units/ml	25 - 45
---------	-----------	---------------	---------

Undetectable CH50 & AH50 activity observed. Please consider to repeat the assay to confirm the findings. If the abnormality persists upon repeated testing, possibility of deficiency in C3 or terminal pathway

Diagnosis: Terminal complement pathway deficiency



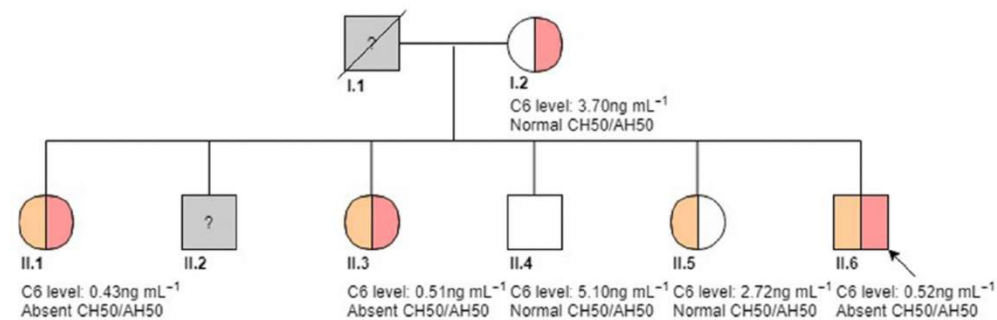
ORIGINAL ARTICLE

Novel pathogenic mutations identified in the first Chinese pedigree of complete C6 deficiency

Philip H Li^{1†}, William WY Wong^{2†}, Evelyn NY Leung², Chak-sing Lau¹ & Elaine Au²

¹Division of Rheumatology & Clinical Immunology, Department of Medicine, Queen Mary Hospital, The University of Hong Kong, Hong Kong

²Division of Clinical Immunology, Department of Pathology, Queen Mary Hospital, Hong Kong



Legend

p.Arg596Ter rs142881576

p.Arg606Ter rs191386155

Wild Type

Unknown

Learning objectives

- **Appreciation** of the clinical features and **when to suspect immunodeficiencies**
- **Understand the approach and first-line diagnostic tests** for workup of suspected immunodeficiencies
- **Be aware** of second-line diagnostic tests and broad classification of primary immunodeficiencies
- **Be able to explain the principles and monitoring of immunoglobulin replacement therapy** for antibody deficiencies

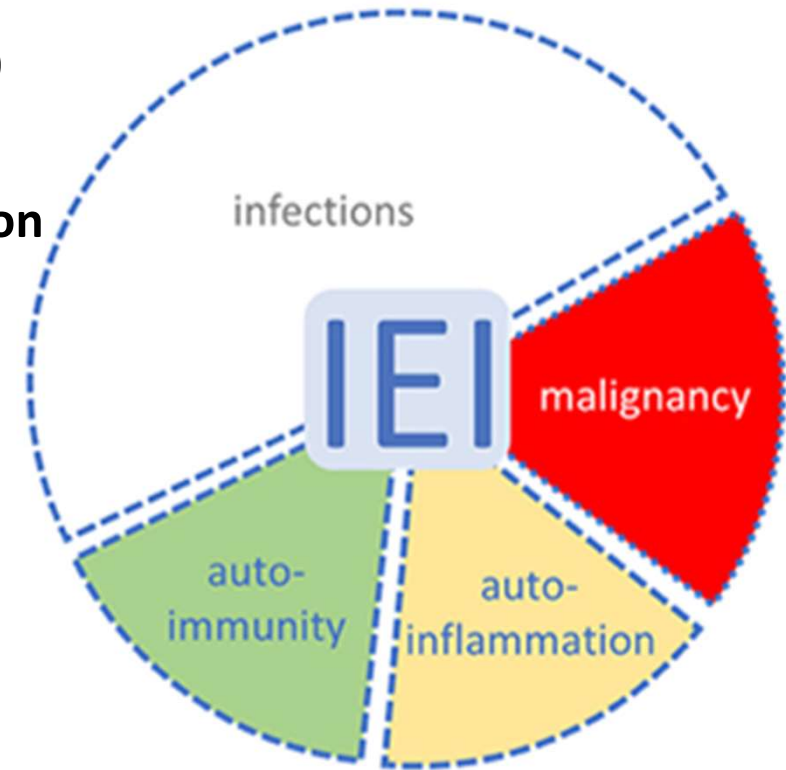


What are 'immunodeficiencies'?

- **Primary** Immunodeficiencies (PIDs) ← **More emphasised & studied!**
 - **Inherent** dysfunctions of the immune system
 - Generally **genetic** in aetiology
 - Can be hereditary or arising from *de novo* mutations
- **Secondary** Immunodeficiencies (SIDs) ← **Much much more common!**
 - Secondary to **other conditions or pathologies**
 - Potential causes include a variety of factors affecting immune function
 - e.g. lymphoproliferative diseases and malignancies, medications, infections, advanced age, malnutrition, protein-losing states....

Primary Immunodeficiencies (PIDs) *aka inborn errors of immunity (IEI)*

- ‘Rare’ inborn errors of immunity (~1:10,000 population)
- Classically have **increased susceptibility to infections**
- Can also be associated with **autoimmunity, inflammation** and increased risk of **malignancy**
- Possible hereditary, and risk **in other family members**
- Management and treatment options vary and can be **severely limited without precise diagnosis**
 - Antimicrobial prophylaxis
 - Immunoglobulin replacement
 - Targeted therapies
 - Hematopoietic stem cell transplant
 - *Gene therapy*
- **Rapid advances with New Generation Sequencing**



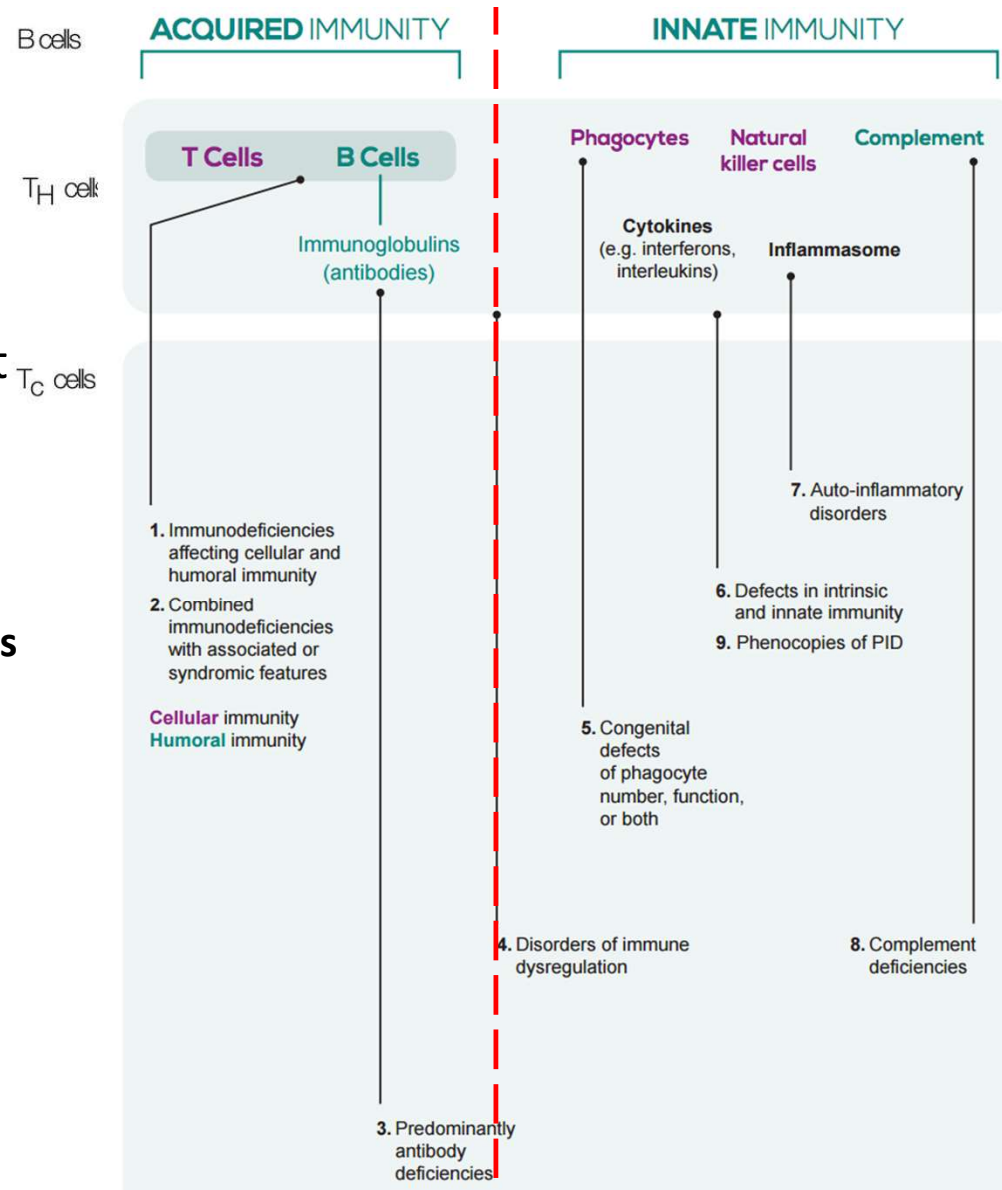
Bomken et al, Front. Immunol., 12 December 2018

Classification of PIDs

> 400 distinct disorders with >430 different gene defects (*and growing!*)

- IUIS PID classification

1. Combined immunodeficiencies (CID)
2. CID with associated or syndromic features
3. Antibody deficiencies
4. Immune dysregulation
5. Phagocytic defects
6. Intrinsic and innate immunity
7. Autoinflammatory disorders
8. Complement deficiencies
9. Bone marrow failure
10. Phenocopies of PID



Role of Immunologist



Screening

- Recognizing potential cases
- Selection of initial investigations and working diagnoses
- Acute management
- Appropriate and timely referral



Diagnosis

- Exclude (first) the reversible, treatable or life-threatening
- Selection of “second line” tests based on clinical presentation, initial investigations and microbiological results



Management

- Replacement therapy
- Anti-microbial prophylaxis
- Treatment of autoimmune, autoinflammatory manifestations
- Disease monitoring and prevention of complications
- ? Transplantation / Gene Rx



Family Screening

- Depending on mode of inheritance
- Offspring, siblings, parents ...
- Preconception counselling
- Prenatal diagnosis
- Early treatment and prevention of complications

General principles

- Not as rare as previously thought – ever-expanding list!
- Many patients remain undiagnosed or present later in life
- **High index of suspicious for early diagnosis and intervention**
- Vigilance – **warning signs of primary immunodeficiencies**
- Differentials: exclude anatomical and functional disorders
- May also present with non-infectious symptoms

Warning signs of immunodef

1. Disproportionately frequent infections
2. Disproportionately severe infections
3. Family history
4. Unusual clinical presentation
5. Unusual microorganisms
6. Unusual sites of infection
7. Clinical phenotype suggestive of syndrome associated with immunodeficiency
8. *Infections after biologicals*

SPUR = **S**erious, **P**ersistent, **U**nusual & **R**ecurring

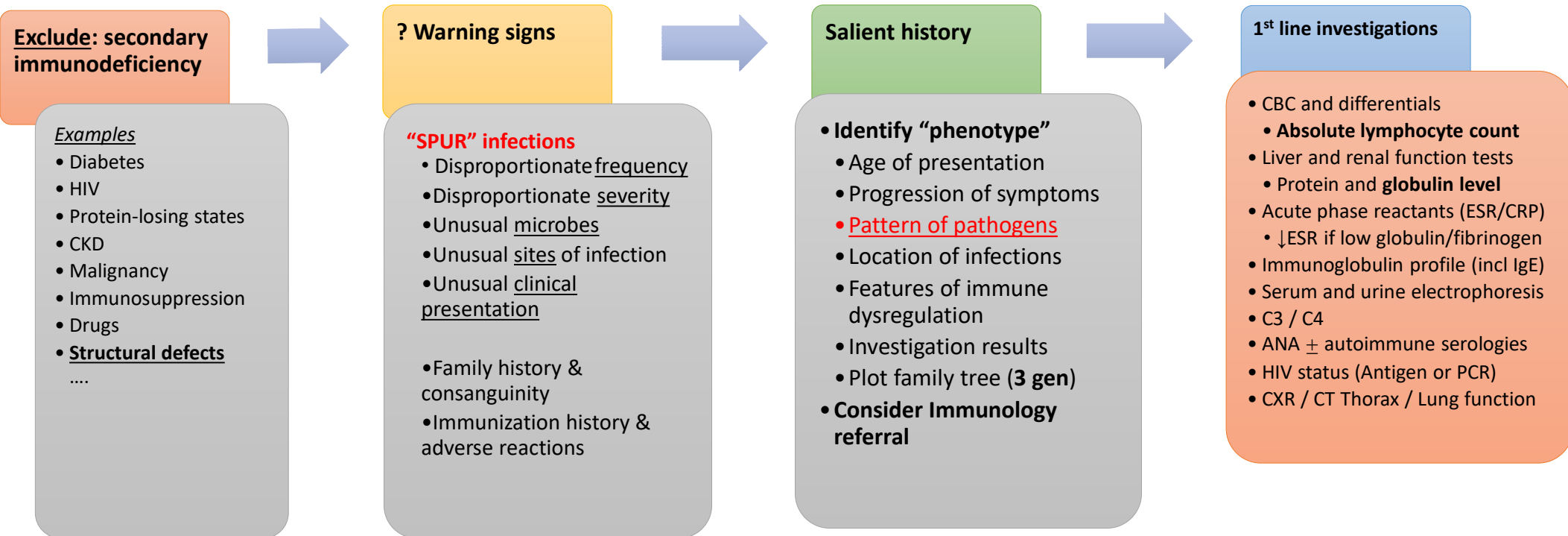
10 Warning Signs Primary Immunodeficiency



Jeffrey Modell
Foundation

Curing PI. Worldwide.

Initial evaluation



Second-line investigations

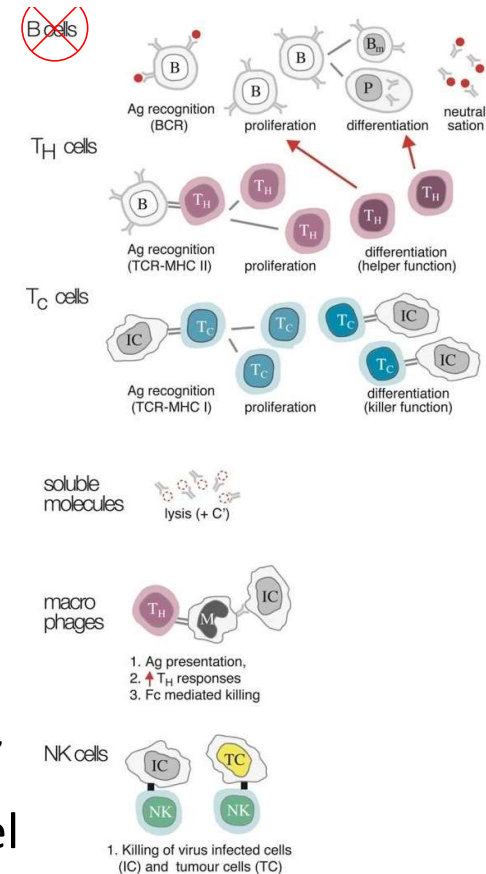
(Based on suspected phenotype)

- Lymphocyte subsets: CD3, CD4, CD8, CD16/56, CD19
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Antibody immunodeficiencies

- **The most common form of primary or secondary immunodeficiency**
- e.g. Agammaglobulinemia, Common Variable Immunodeficiency, B-cell depleting therapies, Chronic lymphocytic leukemia, Multiple Myeloma
- Recurrent sinopulmonary (“idiopathic” bronchiectasis) and gastrointestinal infections (chronic diarrhea)
 esp. **Encapsulated bacteria** (*Haemophilus influenzae*, *Streptococcus pneumoniae*, *Neisseria meningitidis*), ***Giardia***, **enteroviruses**
- Autoimmune cytopenias, interstitial lung disease, inflammatory bowel disease
- **Treatment:** Ig replacement, prophylactic antibiotics



Philip H. Li*, Chak-Sing Lau

Secondary antibody deficiency and immunoglobulin replacement

Table 1: Common examples of drug-induced hypogammaglobulinemia

Anti-epileptics	Immunosuppressants	Monoclonal antibodies	Others
Carbamazepine	Azathioprine	Belimumab	Antimalarial agents
Lamotrigine	Cyclophosphamide	Ofatumumab	Captopril
Levetiracetam	Cyclosporine	Rituximab	Fenclofenac
Oxcarbazepine	Gold salts	Imatinib	
Phenytoin	Mycophenolate		
Valproate	Penicillamine		
	Steroids		
	Sulphasalazine		

Box 1: Recommendations on screening for secondary hypogammaglobulinemia**When to screen for hypogammaglobulinemia:**

- History of infection with unusual severity or frequency, or lack of response to therapy
- History of characteristic pathogens (e.g., encapsulated bacteria or severe/persistent viral infections)
- Before commencement of any immunosuppressive therapies
- Every 3-6 months while on “stronger” immunosuppression (such as cytotoxic chemotherapy, anti-CD20), and until at least 6-12 months after last dose of anti-CD20 therapy

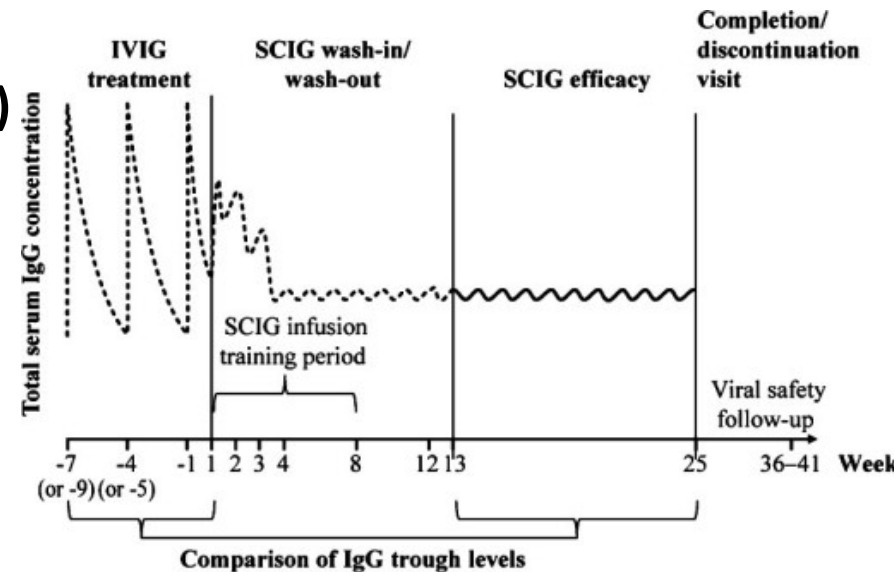
Immunoglobulin therapy

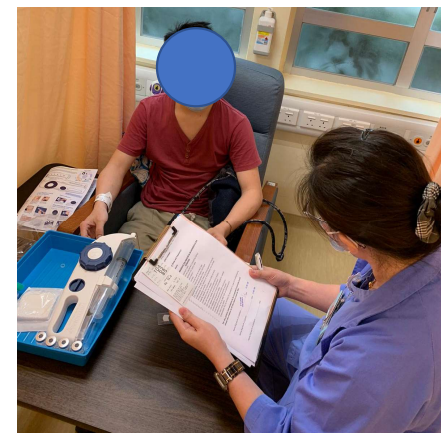
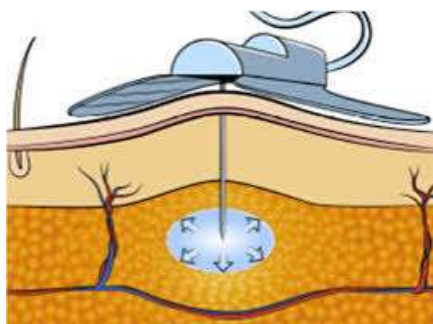
- Immunoglobulin (Ig) derived from **pool of plasma donors (>1000)**
- **Mostly IgG only (negligible other immunoglobulin isotypes)**
- Routes: **IVIG** vs **SCIG** vs **IMIG**
- Specific types of Ig: Hyperimmune globulin (specific Ab), Anti-D Ig
- Main indications
 1. **Ig replacement** – protection against infection
 - Primary or secondary antibody deficiencies
 - ~0.4-0.5g/kg per month
 2. **Immunomodulatory** - suppression of inflammation
 - Neurology, rheumatology, haematology, renal, transplantation, infections...
 - ~1-2g/kg per dose

! Most serology tests become inaccurate after Ig replacement

IVIg vs SCIg

- IV: Once per month
- SC: Once per week, daily push, *facilitated SCIg*
 - **More physiological:** Avoids trough levels
 - **More convenient:** home therapy, fewer admissions (↓infection risk)
 - **Easier:** Does not require venous access
 - **Safer:** Fewer systemic adverse reactions
- **Cost saving (for patients and healthcare costs!)**
 - Lower episodes of infections
 - Fewer admissions / bed occupancy
 - Save manpower for iv administration
 - Fewer follow-up visits and monitoring
 - Less absenteeism and improved productivity
 - Improved quality of life





A Comparative Study of Intravenous Immunoglobulin and Subcutaneous Immunoglobulin in Adult Patients with Primary Immunodeficiency Diseases: A Systematic Review and Meta-Analysis

Hosein Shabaninejad, Asra Asgharzadeh, Nima Rezaei & Aziz Rezapoor

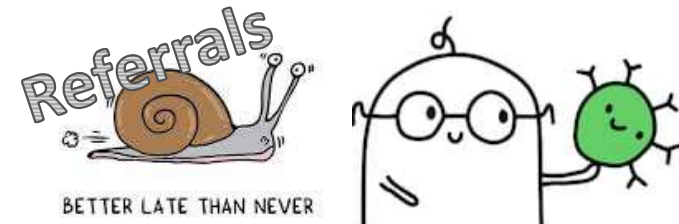
To cite this article: Hosein Shabaninejad, Asra Asgharzadeh, Nima Rezaei & Aziz Rezapoor (2016): A Comparative Study of Intravenous Immunoglobulin and Subcutaneous Immunoglobulin in Adult Patients with Primary Immunodeficiency Diseases: A Systematic Review and Meta-Analysis, Expert Review of Clinical Immunology, DOI: [10.1586/1744666X.2016.1155452](https://doi.org/10.1586/1744666X.2016.1155452)



SCIG leads to:
higher level of immunoglobulin and a
reduction in side effects
~same infection rate as IVIG

Take home messages

- Primary immunodeficiencies are not as “rare” as you think
? “**Never encountered**” vs. **Never diagnosed**
- **Vigilance** for potential cases, or easily overlooked!
- **Exclude secondary causes** first, then consider **disease phenotype**
(*pattern of pathogens, types of infection or immunodysregulation, age of onset, gender, family history, prelim investigations etc.*)
- **First-line ± second-line investigations** if indicated
- Consider **timely referral or consultation!**



Questions?

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